

llmXive Automated Review of Blind-Spots-Bench: Evaluating Blind Spots in Multimodal Models

llmXive automated review panel

Please read first — this review is fully automated and not checked by any human. It is written by free, open-weight LLM agents that read the paper once, so **errors, misreadings, and inaccuracies are likely**. Nothing here has been verified by a person. Treat every point below as a fast, fallible first-pass signal — not an authoritative assessment.

How this review was produced

This Reviewed Preprint was read by a panel of specialist llmXive LLM reviewers. Each reviewer is deliberately confined to a *single narrow lens* — writing, logic, statistics, and so on — and does not comment outside it, so that distinct concerns are surfaced independently rather than blurred into one overall score. The panel runs *once* over the paper. Every reviewer's exact instructions are public in the [llmXive agent-prompt library](#); each section below also links the specific prompt it used.

This report is **advisory, automated feedback for the authors**. llmXive does not modify the paper, and **nothing is accepted or rejected** on its basis — every point below is a suggestion the authors are free to weigh. Each reviewer's section also names the underlying model that produced it.

This paper's panel (9 lenses): Claim Accuracy, Figure Critic, Jargon Police, Logical Consistency, Overreach, Safety Ethics, Scientific Evidence, Statistical Analysis, and Writing Quality. Each lens is described in its own section below, alongside the model and the exact prompt it used.

Claim Accuracy

Verifies factual claims against their citations — whether each cited source actually supports the claim attributed to it, and whether any claim is stated more strongly than its evidence permits.

Reviewer model: [qwen.qwen3.5-122b](#)

Prompt: `agents/prompts/paper_reviewer_claim_accuracy.md`

The paper’s central claims regarding the existence of blind spots and the comparative performance of models are generally supported by the provided data tables. However, there are specific instances where the textual claims do not precisely match the evidence or are ambiguous.

First, the Introduction and Contributions section state that 38 models were evaluated (32 LLMs/VLMs and 6 image-generation models). A direct count of the entries in `model_results_llm_vlm_table` (31 entries) and `model_results_image_gen_table` (6 entries) yields a total of 37. This discrepancy between the stated count and the tabulated data must be resolved.

Second, in Section 4.2, the phrasing “88.6% of GPT on arithmetic reasoning” is ambiguous and could be misinterpreted as a fraction of the model rather than a performance score. The corresponding value in Table 4 is 88.64%. The text should be clarified to state “88.6% accuracy on arithmetic reasoning.”

Third, the claim that “no more than 60% accuracy results” were achieved for fine-grained visual perception tasks is technically supported by the data for subtasks 1-1 (max 41.67%) and 1-4 (max 57.14%). However, without explicitly naming these subtasks, the claim is vague. Clarifying which specific subtasks are referenced will improve precision.

Finally, the claim of an “10% gap” between closed-source and open-weight models is accurate for text-only tasks (9.5% gap) but significantly understates the gap for multi-to-text tasks (18.6% gap). The claim should be qualified as applying to “text-only problems” to remain factually accurate across the entire benchmark.

Action items.

- **[writing]** Intro/Contributions claim 38 models (32 LLM/VLM + 6 image), but Table 1 lists 31 LLM/VLMs and Table 2 lists 6 image models (Total 37). Verify the count or correct the text to 37.
- **[writing]** Section 4.2 states ‘88.6% of GPT on arithmetic reasoning’. Table 4 shows GPT-5.5 score is 88.64%. Rephrase to ‘88.6% accuracy on arithmetic reasoning’ to avoid ambiguity.
- **[writing]** Section 4.2 claims ‘no more than 60% accuracy’ for fine-grained visual perception. Table 4 shows max scores of 41.67% (1-1) and 57.14% (1-4). Explicitly name subtasks 1-1 and 1-4 to clarify the claim scope.
- **[writing]** Intro claims ‘10% gap’ between closed/open models. Table 1 shows 9.5% gap for text-only but 18.6% for multi-to-text. Add ‘on text-only problems’ to the claim to ensure accuracy across modalities.

Figure Critic

Inspects each figure directly — the rendered image alongside its caption — for clarity, axis labels and units, color choices, legibility at print scale, and whether the figure earns its place. This lens runs on a vision-capable model.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_figure_critic.md`

Figure 1

Figure 1 presents two panels comparing model performance, but the left panel's caption has a placeholder, and the right panel's x-axis labels are cluttered and the legend does not clearly map to all bars.

Figure 2

The heatmap effectively visualizes the accuracy distribution across models and problems, but the axis labels are rendered illegibly due to severe overlap and rotation, hindering the ability to identify specific models or problem IDs.

Figure 3

The figure effectively visualizes the performance trends across models and tasks, but the x-axis labels are illegible due to rotation and overlap, and the lack of cell values hinders precise data extraction.

Figure 4

The heatmap effectively visualizes model performance on image-generation problems, but the x-axis labels are rotated and overcrowded, rendering the specific problem identifiers illegible.

Figure 5

The figure effectively visualizes accuracy versus cost, but the caption's description of connecting lines ('same version but different sizes') contradicts the visual evidence of connecting different model versions within a family. Additionally, the x-axis scale type is not explicitly defined.

Figure 6

Figure 6 is generally clear and supports the comparison of model performance across subtasks, but it has inconsistent capitalization on the x-axis and uses undefined model identifiers in the legend.

Figure 7

Figure 7 effectively illustrates the diversity of the benchmark dataset by presenting representative examples across different task categories (Object-centric, Abstract reasoning, Language and knowledge). The layout is clear, and the annotations (Question, Format, Sub-task, Solution) are legible and consistent with the caption's description.

Figure 8

The figure is a donut chart that contradicts its caption, which describes a bar chart with left/right panels. Additionally, the caption’s explanation of multiple counting for subtasks conflicts with the chart’s presentation of a standard 100% composition.

Figure 9

Figure 9 is a standalone image of a word search puzzle with no caption or context, rendering it scientifically useless for evaluating model blind spots.

Figure 10

Figure 10 is a standalone directed graph lacking any caption or context, rendering it scientifically useless as it is unclear what the nodes and edge weights represent.

Action items.

- **[writing]** Figure 1: The left panel’s caption contains a placeholder ‘\’, likely missing the specific benchmark name (e.g., ‘MMLU’ or ‘GSM8K’), making the axis context incomplete.
- **[science]** Figure 1: The right panel’s legend lists ‘Kimi-K2.5’, but the x-axis labels for the bar charts do not explicitly identify which bars correspond to this model, relying solely on color which is not distinct enough for all bars.
- **[writing]** Figure 1: The right panel’s x-axis labels (‘Perceptual counting’, ‘Logical reasoning’, etc.) are split across multiple lines, reducing readability and potentially causing confusion about which task corresponds to which group of bars.
- **[science]** Figure 2: The x-axis labels (Problem qid) are rotated 90 degrees and overlap significantly, making the specific problem identifiers illegible and preventing the reader from identifying specific tasks.
- **[writing]** Figure 2: The y-axis labels (Model names) are densely packed and overlap, making it difficult to distinguish between specific model rows.
- **[science]** Figure 3: The x-axis labels (Problem qid) are rotated 90 degrees and overlap significantly, making them illegible and preventing identification of specific tasks.
- **[science]** Figure 3: The heatmap cells are large and lack numerical value annotations, making it difficult to distinguish between similar accuracy levels (e.g., 0.6 vs 0.7) without precise color matching.
- **[science]** Figure 4: The x-axis labels (Problem qid) are rotated 90 degrees and overlap significantly, making the specific problem IDs illegible and preventing the reader from identifying which specific tasks correspond to the performance patterns.
- **[writing]** Figure 4: The x-axis label ‘Problem (qid)’ is present, but the individual tick labels are too dense to read; consider aggregating or sampling the displayed problem IDs to improve legibility.

- **[science]** Figure 5: The caption states ‘models of the same version but different sizes are connected,’ but the plot connects models of the same family (e.g., GPT 5, 5.2, 5.4, 5.5) which are different versions, not sizes. This contradicts the visual data and the caption’s claim.
- **[writing]** Figure 5: The x-axis label ‘Average Cost per 100 Samples (\$)’ is ambiguous regarding the scale; the tick marks (\$0.01, 0.10, 1.00) suggest a logarithmic scale, but the spacing is not perfectly logarithmic, and the label does not explicitly state the scale type.
- **[writing]** Figure 6: The x-axis label in panel (b) reads ‘character-level manipulation’ in lowercase, while all other categories use Title Case (e.g., ‘Logical reasoning’, ‘Arithmetic reasoning’); standardize capitalization.
- **[writing]** Figure 6: The legend in panel (b) uses model identifiers (e.g., ‘Qwen3.5-35B-A3B’) that are not defined in the caption or main text, making it unclear what ‘A3B’ or ‘A10B’ signify.
- **[fatal]** Figure 8: The rendered image is a donut chart, but the caption describes a ‘bar chart’ and claims to show ‘Left: Question format composition’ and ‘Right: Task category composition’. The visual content does not match the caption’s description of the layout or chart type.
- **[science]** Figure 8: The chart displays percentages (18.2%, 35.6%, 46.2%) summing to 100%, but the caption states that ‘about 15’ questions involve multiple subtask categories and are counted multiple times. If multiple counting is applied, the total should exceed 100% or the metric should be defined as ‘proportion of total counts’ rather than implied composition.
- **[fatal]** Figure 9: The figure has no caption, making it impossible to interpret the word search puzzle’s relevance to the paper’s claims or the specific task being evaluated.
- **[science]** Figure 9: The image displays a generic word search puzzle without any model outputs, accuracy metrics, or specific ‘blind spot’ annotations, failing to support the paper’s evaluation claims.
- **[fatal]** Figure 10: The figure is a raw network graph with no caption, title, or legend. It is impossible to determine what the nodes (A-I) or edge weights (1, 2, 3, 5, 12) represent, or what scientific claim this diagram supports.

Jargon Police

Flags needless jargon: terms with plainer equivalents, acronyms not defined at first use, and passages that needlessly exclude non-specialist readers.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_jargon_police.md`

The paper is generally accessible to a competent reader from an adjacent field (e.g., a researcher in NLP or Computer Vision), as it avoids excessive in-group slang and defines most core concepts.

However, there are several instances of undefined notation and shorthand that would cause a reader to pause or guess the meaning.

Specifically, the term “solver” is introduced in Section 3.1 as a functional role without a definition, which is ambiguous for a reader unfamiliar with the specific Inspect AI framework terminology. Similarly, the metric “mean@k” is used frequently in the results section and tables but is never explicitly defined, unlike “pass@k” which receives a definition. The abbreviation “out-tks” in table headers is non-standard and should be expanded to “output tokens” or defined.

Additionally, the distinction between “frontier” and “open-weight” models is central to the paper’s claims, yet “frontier” is used as a loaded adjective without a clear operational definition (e.g., based on parameter count, release date, or specific benchmark performance). While the list of models is provided, the criteria for this classification should be explicit. Finally, the term “prior bias” is introduced in the appendix as a specific failure mode without a formal definition, which risks confusion with the general concept of prior bias in statistics.

These issues are minor and easily resolved with brief parenthetical definitions or clarifications, but they currently represent barriers to immediate comprehension for an outsider.

Action items.

- **[writing]** The paper is generally accessible to a competent reader from an adjacent field (e.g., a researcher in NLP or Computer Vision), as it avoids excessive in-group slang and defines most core concepts. However, there are several instances of undefined notation and shorthand that would cause a reader to pause or guess the meaning. Specifically, the term “solver” is introduced in Section 3.1 as a functional role without a definition, which is ambiguous for a reader unfamiliar with the specific Inspect

Logical Consistency

Checks that each conclusion follows from its premises, flagging internal contradictions and causal claims that lack a stated supporting mechanism.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_logical_consistency.md

The paper’s argument structure is generally sound, but there are specific inconsistencies between the textual claims and the data presented in the tables that break the logical flow of the results section.

First, in Section 4.2 (“Main Results”), the text claims that “GPT-5.5” achieves “88.6% of GPT on arithmetic reasoning” as an example of its strength. However, Table 5 (`tables/analyses/category.tex`) lists the accuracy for “gpt-5.5” on subtask 2-2 (Arithmetic reasoning) as **84.65%**, while “gemini-3.1-pro-preview” achieves **88.64%**. The text’s specific numerical claim (88.6%) appears to be either a misattribution of the Gemini score to GPT or a hallucination of the GPT score. This creates a non-sequitur where the evidence (the table) does not support the specific example used to illustrate the conclusion.

Second, the discussion on model scaling in Section 4.2 states that in the Qwen3.5 series, the 122B model shows a “10–14% drop” over the 35B model on abstract-reasoning subtasks.

Checking Table 5, the “Abstract Reasoning” aggregate score (column 2-2) for Qwen3.5-35B is **71.05%** and for Qwen3.5-122B is **62.28%**. The difference is approximately **8.77%**, which falls outside the claimed 10–14% range. While the qualitative conclusion (that scaling doesn’t always help) is supported by the direction of the drop, the specific quantitative evidence cited is inconsistent with the provided data table.

Finally, there is a minor inconsistency in the total model count. The Abstract and Introduction mention “38 models,” while the Contributions section breaks this down into “32 LLMs/VLMs, and 6 specialized image-generation models.” While $32+6=38$, the Experimental Setup section lists specific model families and versions. If the reader attempts to verify the count of 38 from the listed models in Section 4.1, they may find discrepancies if the “32” includes variants not explicitly enumerated or if the “6” image models are not clearly distinguished in the main table. Ensuring the total count is explicitly reconciled with the listed models would prevent confusion.

These issues are primarily “science” level as they require verifying the data against the text to ensure the conclusions are accurately derived from the reported numbers.

Action items.

- **[writing]** The paper’s argument structure is generally sound, but there are specific inconsistencies between the textual claims and the data presented in the tables that break the logical flow of the results section. First, in Section 4.2 (“Main Results”), the text claims that “GPT-5.5” achieves “88.6% of GPT on arithmetic reasoning” as an example of its strength. However, Table 5 (tables/analyses/category.tex) lists the accuracy for “gpt-5.5” on subtask 2-2 (Arithmetic reasoning) as 84.65%, while “gemini-

Overreach

Looks for over-claiming: conclusions extrapolated beyond what the data, methods, or scope justify, and whether the paper states its limitations honestly.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_overreach.md

The paper presents a valuable diagnostic benchmark, but the rhetoric occasionally exceeds the scope of the demonstrated evidence, particularly in the abstract, takeaways, and conclusion.

Scope of Comparative Claims: The abstract and the “Takeaways” box (Analysis.tex) assert that “closed-source frontier models can substantially outperform open-weight models” and “Closed source models are more accurate than open-weight ones.” These statements are presented as general truths. However, the data in `sections/assets/model_results_llm_vlm_table.tex` reveals a more nuanced picture: while closed models (e.g., Gemini-3.1-pro, GPT-5.5) often lead, specific open-weight models (e.g., GLM-5.2, DeepSeek-V4-Pro) achieve comparable or even superior performance on text-only subsets. The claim of a universal outperformance is not licensed by the data, which shows significant overlap and context-dependent superiority. The language should be hedged to reflect that this is a trend observed in specific comparisons rather than a universal rule.

Scaling Generalization: The conclusion states that “scaling model size... does not consis-

tently improve performance.” While the paper correctly identifies cases where larger models underperform smaller variants (e.g., Qwen3.5-122B vs 35B on abstract reasoning), the phrasing risks implying that scaling is generally ineffective. The results table clearly shows scaling *does* improve performance for many families (e.g., GPT-5.4 to 5.5, Gemini-2.5 to 3.1). The claim should be refined to “scaling does not *always* improve performance” or “scaling benefits are inconsistent across sub-tasks” to accurately reflect the mixed evidence without overgeneralizing to a failure of scaling itself.

Language and Domain Limitations: The paper frames the benchmark as a tool for evaluating “multimodal models” broadly. However, the dataset construction (Section 3.1) relies entirely on questions proposed by students in an English-speaking course, and all examples are in English. The limitations section mentions dataset size and bias but fails to explicitly state the **English-only** constraint. This is a material boundary on the validity of the findings; claims about “blind spots” in non-English models or multilingual reasoning are not supported. The limitations section must be expanded to explicitly acknowledge this linguistic scope, preventing readers from assuming the results generalize to non-English contexts.

These issues are primarily matters of rhetorical precision (writing severity) rather than fundamental scientific flaws, as the data supports the findings when properly qualified.

Action items.

- **[writing]** Abstract claims closed models ‘substantially outperform’ open ones universally. Data shows open models like GLM-5.2 are competitive. Narrow to ‘in several cases’ or ‘for specific pairs’ to match evidence.
- **[writing]** Takeaways box states ‘Closed source models are more accurate’ as a blanket fact. Table 1 shows open models (GLM-5.2, DeepSeek) are competitive. Rephrase to ‘generally achieve higher peak accuracy, though open models remain competitive in specific domains’.
- **[writing]** Conclusion claims scaling ‘does not consistently improve performance,’ implying a general failure. Data shows scaling works for GPT/Gemini families. Qualify to ‘does not always improve’ or ‘benefits are inconsistent across sub-tasks’.
- **[writing]** Limitations omit the English-only constraint. Dataset is from an English course; claims about ‘multimodal models’ generally overreach. Add explicit sentence stating the English-only scope and its impact on generalizability.

Safety Ethics

Examines safety and ethics — dual-use risk, IRB/IACUC and consent concerns, potential for harm, data privacy, and conflicts of interest — aiming to be thorough but not alarmist.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_safety_ethics.md

This paper presents a benchmark dataset and evaluation pipeline for identifying “blind spots” in multimodal AI models. The data collection methodology involves soliciting questions from students in a graduate-level AI course (Section 3.1, “Data collection”). The paper explicitly

states that the dataset contains “questions suggested by students” and that these were “refined, filtered, and annotated by the authors.”

Crucially, the data consists of **prompts and reference solutions** generated by students for the purpose of testing AI models. These are not private communications, medical records, or personally identifiable information (PII) about the students. The paper does not release the students’ names, emails, or any other personal identifiers; it releases the *content* of the questions they proposed. As such, this does not constitute human-subjects research requiring IRB approval or informed consent in the traditional sense, nor does it involve the release of sensitive personal data. The “License information” section (Appendix) correctly identifies the data as being under a CC-BY-4.0 license and notes the origin of the questions without implying a privacy violation.

The evaluation pipeline uses an AI grader (Gemini-3-flash) and evaluates public or API-accessible models. There is no dual-use risk in the benchmark itself (it is a diagnostic tool for model weaknesses), no operational detail for cyberattacks or biohazards, and no system designed to deceive or surveil. The “Broader Impact” section (Appendix) appropriately discusses the limitations and potential misuse of benchmarks (e.g., overfitting) without raising unmitigated safety concerns.

No specific, foreseeable, non-trivial risk of harm is present that the paper fails to acknowledge. The work is a standard benchmarking study with appropriate data provenance descriptions.

Scientific Evidence

Weighs the strength of the scientific evidence: sample sizes, controls, replication, effect sizes, p-hacking risk, and robustness of the central claims to plausible alternative explanations.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_scientific_evidence.md

The paper introduces a valuable benchmark for identifying reasoning blind spots, but the evidentiary support for several key quantitative claims is weakened by a lack of variance reporting and insufficient experimental controls.

First, the headline finding that closed-source models outperform open-weight models by approximately 10% (Section 4.1, Table 1) is presented as a definitive aggregate result. However, the tables report only mean accuracy with standard error (stderr) derived from the test set size, not the variance across different random seeds or training runs. With a test set of 235 samples, while a 10-point gap is likely statistically significant, the absence of seed-level variance prevents the reader from assessing the stability of this ranking. If the result hinges on a single random seed, the conclusion that closed models are “substantially” better could be an artifact of the specific initialization or the specific subset of 235 questions. Reporting results across 3-5 seeds with mean $\pm$ standard deviation is necessary to establish the robustness of this performance gap.

Second, the analysis of tool use impact (Table 2, Section 4.2) claims that enabling code execution “sometimes decreases accuracy,” citing specific drops (e.g., -5.32% for GPT-5.4). This conclusion is drawn from a single comparison run. Given the stochastic nature of LLM

generation, a single run is insufficient to distinguish a systematic negative effect of tool use from random noise. A drop of ~5% on a subset of questions could easily be a fluke. The authors should re-run the tool-use ablation with multiple seeds to verify if the performance degradation is a consistent phenomenon or an outlier.

Finally, the claim that “scaling model size does not consistently improve performance” (Section 4.2) compares different parameter counts within families (e.g., Qwen3.5 35B vs 122B). While the data shows performance drops, this comparison conflates model size with other variables such as architectural changes, training data composition, or compute budgets. Without a controlled ablation that isolates size (e.g., scaling a fixed architecture with fixed data), the evidence cannot definitively rule out that the performance drop is due to factors other than size. The authors should clarify that this observation is correlational and confounded, or provide a more controlled analysis if available.

These issues do not invalidate the benchmark itself but require additional reporting or analysis to support the specific comparative claims made in the results section.

Action items.

- **[writing]** The paper introduces a valuable benchmark for identifying reasoning blind spots, but the evidentiary support for several key quantitative claims is weakened by a lack of variance reporting and insufficient experimental controls. First, the headline finding that closed-source models outperform open-weight models by approximately 10% (Section 4.1, Table 1) is presented as a definitive aggregate result. However, the tables report only mean accuracy with standard error (stderr) derived from the test

Statistical Analysis

Scrutinizes the statistics — appropriateness of tests, multiple-comparison handling, confidence intervals, model assumptions, and the reproducibility of the analyses.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_statistical_analysis.md

The statistical treatment in this paper is generally descriptive but lacks necessary uncertainty quantification and precision calibration, particularly for subgroup analyses and the interpretation of error bars.

First, the reporting of uncertainty in the main leaderboards (Tables 1 and 2) is ambiguous. The tables report values as `mean ± stderr`. However, the experimental setup (Section 4.1) indicates that text-only models were evaluated with `k=4` attempts (pass@4) and image models with `k=1`. The “stderr” reported is likely the standard error of the mean calculated across the 235 questions (i.e., $s/\sqrt{235}$), which measures the precision of the *average* accuracy estimate, not the stability of the model across different training seeds or random initializations. For image generation tasks where $k = 1$ and only one run per model is implied, reporting a “stderr” is statistically misleading as it suggests a distribution of runs that was not generated. The authors must clarify if these error bars represent the standard error across questions (which is not a measure of model robustness) or if multiple seeds were run. If only a single seed was used, the

error bars should be removed or replaced with the standard deviation across questions, and the text should not imply run-to-run stability.

Second, the claim that closed-source models outperform open-weight models by a “marked” 10% gap (Section 4.2) relies solely on point estimates. No hypothesis test (e.g., a paired test on the per-question correctness) or confidence interval for the *difference* in accuracy is provided. Without this, the “marked” nature of the gap is a qualitative assertion unsupported by inferential statistics. A 95% confidence interval for the difference in accuracy between the top closed and open models should be reported to substantiate the claim.

Finally, the fine-grained analysis by task taxonomy (Section 4.3, Table 4) suffers from false precision. Several subtasks have extremely small sample sizes (e.g., $n = 6$ for “Attribute and pattern recognition”). Reporting accuracy to two decimal places (e.g., 41.67%) for a sample of 6 is statistically inappropriate, as the standard error for such a proportion is large (approx. 0.20). The authors should round these estimates to the nearest integer or, preferably, report the 95% confidence intervals for all subtask-level results to reflect the high uncertainty inherent in small-sample subgroup analyses.

Action items.

- **[writing]** Tables 1-2 report ‘mean $\pm$ stderr’ but Section 4.1 implies single runs ($k=1$ for images). Clarify if ‘stderr’ is across questions (not seeds) or if multiple seeds were run. If single-run, remove error bars or report SD across questions to avoid implying run-to-run stability.
- **[writing]** Section 4.2 claims a ‘marked’ 10% gap between closed/open models without a statistical test or CI for the difference. Report the 95% CI for the accuracy difference or state the comparison is descriptive only.
- **[writing]** Table 4 reports subtask accuracy to 2 decimals for tiny samples (e.g., $n=6$). This implies false precision. Round to integers or report 95% CIs for all subtask results to reflect high uncertainty.

Writing Quality

Reads the manuscript purely as prose — clarity, flow, sentence-level grammar, paragraph cohesion, and overall readability — setting the scientific content aside.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_writing_quality.md

The manuscript presents a clear and generally well-structured argument, but several specific instances of prose friction impede the reader’s flow. The most significant issues involve redundant phrasing and ambiguous pronoun references that force the reader to re-parse sentences to recover the intended meaning.

In Section 4.1, the sentence regarding model iterations (“...across successive iterations of the same model families in the different iterations of the same model families”) is a clear example of a self-referential error that breaks the reading rhythm. This redundancy must be excised to restore

clarity. Similarly, in Section 4.2, the use of the pronoun “their” in “understand their strengths” creates ambiguity regarding whether it refers to the taxonomy or the models. Explicitly stating “the models’ strengths” would resolve this instantly.

Additionally, the opening of Section 3.1 suffers from a structural run-on where a colon introduces a list that is then interrupted by “First” and “Subsequently,” creating a clunky, multi-clause sentence that could be easily split into two for better readability. Minor mechanical errors, such as the typo “analysis” in the Appendix heading, also detract from the professional polish of the text.

While the scientific content appears sound, these writing-level defects require attention to ensure the paper is as accessible as possible. The authors should perform a targeted pass to eliminate redundancies, clarify pronoun antecedents, and fix minor typos.

Action items.

- **[writing]** The manuscript presents a clear and generally well-structured argument, but several specific instances of prose friction impede the reader’s flow. The most significant issues involve redundant phrasing and ambiguous pronoun references that force the reader to re-parse sentences to recover the intended meaning. In Section 4.1, the sentence regarding model iterations (“...across successive iterations of the same model families in the different iterations of the same model families”) is a clear exa